

Module 3: Complex Integration

Taylor's series → Let $f(z)$ be analytic on $D \subset \mathbb{C}$ & $z_0 \in D$, then the Taylor's series of the function f about z_0 is given by

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n = a_0 + a_1 (z - z_0) + a_2 (z - z_0)^2 + \dots \quad \text{--- ①}$$

where $a_0 = f(z_0)$ & $a_n = \frac{f^{(n)}(z_0)}{n!}$, $n \geq 1$.

are called the coefficients of Taylor's series ①.

Q1 Find the Taylor's series of $f(z) = e^z$ about $z_0 = 2$.

Solⁿ

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

$$e^z = \sum_{n=0}^{\infty} a_n (z - 2)^n \quad \text{--- ①}$$

$$a_0 = f(2) = e^2 \quad \& \quad a_n = \frac{f^{(n)}(2)}{n!}, \quad n \geq 1$$

$$\therefore f^{(n)}(z) = e^z$$

$$\therefore f^{(n)}(2) = e^2$$

hence $a_n = \frac{e^2}{n!}$, for $n \geq 1$

From ①, $e^z = \sum_{n=0}^{\infty} \frac{e^2}{n!} (z-2)^n$

$$\left[e^z = e^2 \sum_{n=0}^{\infty} \frac{(z-2)^n}{n!} \right]$$

The Taylor's series of $f(z) = e^z$ about $z_0 = 0$ is

$$\left[f(z) = e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} \right]$$

Taylor's series of rational functions about origin $\rightarrow$

Q. Find the Taylor's series of $f(z) = \frac{1}{(z-1)(z+2)}$ about $z=0$,
& determine the region of validity.

Solⁿ

$$f(z) = \frac{1}{(z-1)(z+2)}$$

$$= \frac{1}{3} \left[\frac{1}{z-1} - \frac{1}{z+2} \right]$$

$$= \frac{1}{3} \left[-\frac{1}{(1-z)} - \frac{1}{2(1+\frac{z}{2})} \right]$$

$$= \frac{1}{3} \left[-(1-z)^{-1} - \frac{1}{2} \left(1+\frac{z}{2}\right)^{-1} \right]$$

For real n , the binomial series

$$(1+z)^n = 1 + nz + \frac{n(n-1)}{2!} z^2 + \frac{n(n-1)(n-2)}{3!} z^3 + \dots$$

converges on $|z| < 1$.

$$= \frac{1}{3} \left[- (1+z+z^2+\dots) - \frac{1}{2} \left\{ 1 - \frac{z}{2} + \left(\frac{z}{2}\right)^2 - \dots \right\} \right]$$

Converges when
 $|z| < 1$ & $\left|\frac{z}{2}\right| < 1$

$$\begin{cases} (1-z)^{-1} = 1+z+z^2+z^3+\dots \\ (1+z)^{-1} = 1-z+z^2-z^3+\dots \end{cases}$$

$$= \frac{1}{3} \left[- \sum_{n=0}^{\infty} z^n - \frac{1}{2} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z}{2}\right)^n \right]$$

Ist series converges on the region $R_1: |z| < 1$

IInd " " " " $R_2: \left|\frac{z}{2}\right| < 1$ or $|z| < 2$.

$\therefore$ the region of the convergence of $f(z)$ is $R_1 \cap R_2: |z| < 1$, which contains $z=0$.

Hence
$$f(z) = \frac{1}{(z-1)(z+2)} = \frac{1}{3} \sum_{n=0}^{\infty} \left[-1 + \left(-\frac{1}{2}\right)^{n+1} \right] z^n, \text{ valid on } |z| < 1.$$

Qs Expand the following functions in Taylor's series about the origin & determine its region of validity in each case:

(a) $f(z) = \frac{1}{(z+2)^2(z-3)}$

(b) $f(z) = \frac{1}{(z^2+1)(z-1)}$

Taylor's series of rational functions about $z \neq 0$

Let us substitute $z - z_0 = u$ in the rational function $f(z)$ & then find the partial fraction of f in terms of u & proceed further.

Qs Find the Taylor's series of the function $f(z) = \frac{1}{(z-1)(z-3)}$ about $z=4$ & determine its region of validity.

Sol's Let $u = z - 4$ & put in $f(z)$.

$$f(z) = \frac{1}{(u+3)(u+1)} = \frac{1}{2} \left[\frac{1}{u+1} - \frac{1}{u+3} \right]$$

$$= \frac{1}{2} \left[\frac{1}{(1+u)} - \frac{1}{3(1+\frac{u}{3})} \right]$$

$$= \frac{1}{2} \left[(1+u)^{-1} - \frac{1}{3} \left(1 + \frac{u}{3} \right)^{-1} \right]$$

$$= \frac{1}{2} \left[(1 - u + u^2 - u^3 + \dots) - \frac{1}{3} \left\{ 1 - \frac{u}{3} + \left(\frac{u}{3} \right)^2 - \left(\frac{u}{3} \right)^3 + \dots \right\} \right]$$

$$f(x) = \frac{1}{2} \left[\sum_{n=0}^{\infty} (-1)^n u^n - \frac{1}{3} \sum_{n=0}^{\infty} (-1)^n \left(\frac{u}{3}\right)^n \right]$$

The series converges on the region $R_1: |u| < 1$

II u u u $R_2: \left| \frac{u}{3} \right| < 1$ or $|u| < 3$

$\therefore$ Region of convergence of $f(z)$ is $R_1 \cap R_2 : |u| < 1$ or $|z-4| < 1$

$$80 \quad f(z) = \frac{1}{2} \sum_{n=0}^{\infty} \left[1 - \frac{1}{3^{n+1}} \right] (-1)^n (z-4)^n.$$

which is valid over the interior of the circle $|z-4| < 1$, whose centre at 4 & radius 1.

Q → Find the Taylor's series of $f(z) = \frac{1}{z^2}$ about $z=i$ & determine its region of validity.

Solⁿ Let $u = z - i$ then $z = u + i$

$$\text{so } f(z) = \frac{1}{(u+i)^2} = -\frac{1}{(1+\frac{u}{i})^2}$$

$$= -\left(1 + \frac{u}{i}\right)^{-2}$$

$$= -\left(1 - iu\right)^{-2}$$

$$= -\sum_{n=0}^{\infty} (n+1)(iu)^n$$

$$f(z) = -\sum_{n=0}^{\infty} (n+1)i^n(z-i)^n$$

converges on the region $R : |iu| < 1$ or $|u| < 1$

$$\text{or } |z-i| < 1.$$

$$(1-x)^{-2} = \sum_{n=0}^{\infty} (n+1)x^n$$

converges for $|x| < 1$

We know that $(1-x)^{-1} = \sum_{n=0}^{\infty} x^n$; converges $|x| < 1$

$$(1-x)^{-2} = \frac{d}{dx} \left(\frac{1}{1-x} \right) = \frac{d}{dx} \left(\sum_{n=0}^{\infty} x^n \right)$$

converges for $|x| < 1$
so term by term
diff.

$$(1-x)^{-2} = \sum_{n=1}^{\infty} nx^{n-1}$$

$$\text{or } \left[(1-x)^{-2} = \sum_{n=0}^{\infty} (n+1)x^n \right]$$

Singularity $\rightarrow$ A singularity (or singular point) of a function is the point at which the function ceases to be analytic.
Ex: If $f(z) = \frac{1}{z-2}$ then $z=2$ is singularity of $f(z)$.

Isolated singularity $\rightarrow$ If $f(z)$ is analytic on some deleted neighborhood $N(z_0) - \{z_0\}$ of $z_0 \in \mathbb{C}$, then z_0 is called an isolated singularity of f .

Isolated singularities of rational functions $\rightarrow$ Let a rational function $f(z) = \frac{\phi(z)}{\psi(z)}$, if $\psi(z_0) = 0$ & $\phi(z_0) \neq 0$ then z_0 is called an isolated singularity of f .

Ex: ① Let $f(z) = \frac{1}{z}$; $\forall z \in \mathbb{C}$. Then f is analytic at all points of $\mathbb{C}$ except at $z=0$, hence $z=0$ is an singularity of f .

② Let $f(z) = \frac{(z+3)}{(z^2+1)}$ is analytic on $\mathbb{C}$ except at $z = \pm i$ & $z = \pm i$ are isolated singularity of f .

Laurent's series about an isolated singularity $\rightarrow$ Let z_0 be an isolated singularity of f . Consider f be analytic on the annulus $A: 0 < r < |z - z_0| < R$. Then Laurent's series of f about z_0 is defined as:

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n = \underbrace{\sum_{n=1}^{\infty} a_{-n} (z - z_0)^{-n}}_{\text{Principal part}} + \underbrace{\sum_{n=0}^{\infty} a_n (z - z_0)^n}_{\text{Analytic part or regular part}} \quad \text{--- (1)}$$

where $\left[a_n = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^{n+1}} dz \right]$

& a_n 's are called its coefficients.

Residue at an isolated singularity $\rightarrow$ The coeff. a_{-1} of the 1st negative power $\frac{1}{(z - z_0)}$ of eqn (1) is called as the $\text{Res}(f; z_0)$ of f at z_0 .

Remarks The residue should be computed at an isolated singularity, which lies within the region of validity of the Laurent's series.

The residue is not defined at an isolated singularity not lying in the validity region.

Note If the principal part of Laurent series contains a finite number of terms, say m , the singularity is called pole of order m of $f(z)$, i.e. $a_{-n} \neq 0, \forall n, n > m$.

Laurent's series of rational functions:

Q1 Find the Laurent's series of $f(z) = \frac{1}{z^2(1-z)}$, valid in the regions (i) $R_1: 0 < |z| < 1$, (ii) $R_2: |z| > 1$, (iii) $R_3: 0 < |z-1| < 1$ & (iv) $R_4: |z-1| > 1$. Also find the residues of f at isolated singularities.

Solⁿ Here $z=0$ & 1 are the isolated singularities of f .

(i) On $R_1: 0 < |z| < 1$.

$$\frac{1}{z^2(1-z)} = \frac{1}{z^2} (1-z)^{-1} = \frac{1}{z^2} (1 + z + z^2 + z^3 + \dots)$$

$$= \frac{1}{z^2} + \frac{1}{z} + 1 + z + z^2 + \dots$$

Since $z=0$ lies within R_1 , we find the $\text{Res}(f; 0) = a_{-1} = 1$.

(ii) On $R_2: |z| > 1: \left|\frac{1}{z}\right| < 1$

$$\frac{1}{z^2(1-z)} = \frac{-1}{z^3(1-\frac{1}{z})} = -\frac{1}{z^3} \left(1 - \frac{1}{z}\right)^{-1}$$

$$= -\frac{1}{z^3} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \dots\right) = -\frac{1}{z^3} - \frac{1}{z^4} - \frac{1}{z^5} - \dots$$

iii) On R_3 : $0 < |z-1| < 1$.

Let $u = z-1$ then $0 < |u| < 1$

$$\& f(z) = \frac{1}{z^2(1-z)} = \frac{1}{(u+1)^2(-u)}$$

$$= -\frac{1}{u} (1+u)^{-2} \quad \text{with } |u| < 1$$

$$= -\frac{1}{u} (1 - 2u + 3u^2 - 4u^3 + \dots)$$

$$f(z) = -\frac{1}{u} + 2 - 3u + 4u^2 - \dots$$

$$f(z) = -\frac{1}{(z-1)} + 2 - 3(z-1) + 4(z-1)^2 - \dots$$

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \dots$$

$$(1+x)^{-2} = 1 - 2x + \frac{(-2)(-3)}{2!} x^2 + \dots$$

$$= 1 - 2x + 3x^2 + \dots$$

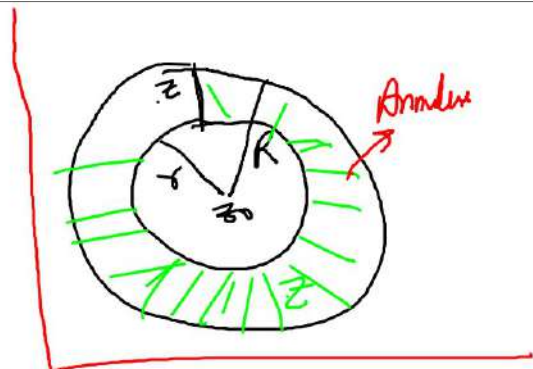
$$= \sum_{n=0}^{\infty} (n+1) (-1)^n x^n$$

$\therefore z=1$ lies within region R_3 , we find the $\text{Res}(f; 1) = -1$.

iv) On R_4 : $|z-1| > 1$.

Let $u = z-1$ then $|u| > 1$ or $|\frac{1}{u}| < 1$

$$\therefore f(z) = \frac{1}{z^2(1-z)} = \frac{1}{(u+1)^2(-u)} = -\frac{1}{u} \cdot \frac{1}{u^2(1+\frac{1}{u})^2} = -\frac{1}{u^3} \left(1 + \frac{1}{u}\right)^{-2}$$



$$= -\frac{1}{u^3} \left(1 - 2 \cdot \frac{1}{u} + 3 \cdot \frac{1}{u^2} - 4 \cdot \frac{1}{u^3} + \dots \right)$$

$$= -\frac{1}{u^3} + \frac{2}{u^4} - \frac{3}{u^5} + \frac{4}{u^6} - \dots$$

$$= -\frac{1}{(z-1)^3} + \frac{2}{(z-1)^4} - \frac{3}{(z-1)^5} + \frac{4}{(z-1)^6} - \dots$$

Q.3 Find the Laurent's series of $f(z) = \frac{1}{(z+1)(z-2)}$, valid in the regions

$$\text{(i) } R_1: 0 < |z+1| < 3$$

$$\text{(ii) } R_2: |z+1| > 3$$

$$\text{(iii) } R_3: 0 < |z-2| < 3$$

$$\text{(iv) } R_4: |z-2| > 3$$

Hence find the residues of f at isolated singularities.

Laurent's series about a point of Analyticity in an Annulus $\Rightarrow$

Qs Find the Laurent series of $f(z) = \frac{1}{z(1-z)}$ on the annulus

A: $1 < |z-2| < 2$.

Sol'n $\because z=0, 1 \notin A$

so we find the Laurent series of f about $z=2$ of analyticity:

Let $u = z-2$, then $1 < |u| < 2$

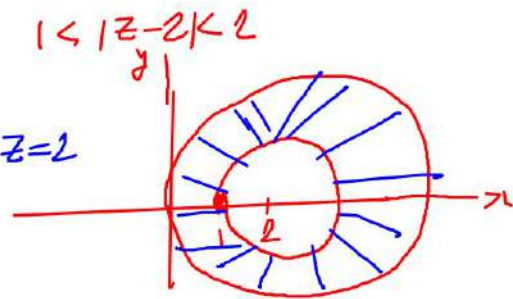
$$\therefore f(z) = \frac{1}{(u+2)(1-u-2)} = \frac{-1}{(u+2)(u+1)}$$

$$= \frac{1}{u+2} - \frac{1}{u+1} \quad ; \quad \left\{ \frac{1}{|u|} < 1 \text{ \& } \frac{|u|}{2} < 1 \right.$$

$$= \frac{1}{2(1+\frac{u}{2})} - \frac{1}{u(1+\frac{1}{u})}$$

$$= \frac{1}{2} \left(1 + \frac{u}{2}\right)^{-1} - \frac{1}{u} \left(1 + \frac{1}{u}\right)^{-1}$$

$$= \frac{1}{2} \sum_{n=0}^{\infty} (-1)^n \left(\frac{u}{2}\right)^n - \frac{1}{u} \sum_{n=0}^{\infty} (-1)^n \left(\frac{1}{u}\right)^n, \text{ putting } u = z-2$$



(2) Singularities of an analytic function

We have already defined a *singular point of a function* as the point at which the function ceases to be analytic.

(i) **Isolated singularity.** If $z = a$ is a singularity of $f(z)$ such that $f(z)$ is analytic at each point in its neighbourhood (i.e., there exists a circle with centre a which has no other singularity), then $z = a$ is called an **isolated singularity**.

In such a case, $f(z)$ can be expanded in a Laurent's series around $z = a$, giving

$$f(z) = a_0 + a_1(z-a) + a_2(z-a)^2 + \dots + a_{-1}(z-a)^{-1} + a_{-2}(z-a)^{-2} + \dots \quad \dots(1)$$

For example, $f(z) = \cot(\pi/z)$ is not analytic where $\tan(\pi/z) = 0$ i.e. at the points $\pi/z = n\pi$ or $z = 1/n$ ($n = 1, 2, 3, \dots$).

Thus $z = 1, 1/2, 1/3, \dots$ are all *isolated singularities* as there is no other singularity in their neighbourhood.

But when n is large, $z = 0$ is such a singularity that there are infinite number of other singularities in its neighbourhood. Thus $z = 0$ is the *non-isolated singularity* of $f(z)$.

(ii) **Removable singularity.** If all the negative powers of $(z-a)$ in (1) are zero, then $f(z) = \sum_{n=0}^{\infty} a_n(z-a)^n$.

Here the singularity can be removed by defining $f(z)$ at $z = a$ in such a way that it becomes analytic at $z = a$. Such a singularity is called a *removable singularity*.

Thus if $\lim_{z \rightarrow a} f(z)$ exists finitely, then $z = a$ is a *removable singularity*.

(iii) **Poles.** If all the negative powers of $(z-a)$ in (i) after the n th are missing, then the singularity at $z = a$ is called a **pole of order n** .

A pole of first order is called a **simple pole**.

(iv) **Essential singularity.** If the number of negative powers of $(z-a)$ in (1) is infinite, then $z = a$ is called an *essential singularity*. In this case, $\lim_{z \rightarrow a} f(z)$ does not exist.

Example 20.30. Find the nature and location of singularities of the following functions:

(i) $\frac{z - \sin z}{z^2}$

(ii) $(z+1) \sin \frac{1}{z-2}$

(iii) $\frac{1}{\cos z - \sin z}$

Solution. (i) Here $z = 0$ is a singularity.

$$\text{Also } \frac{z - \sin z}{z^2} = \frac{1}{z^2} \left\{ z - \left(z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \dots \right) \right\} = \frac{z}{3!} - \frac{z^3}{5!} + \frac{z^5}{7!} - \dots$$

Since there are no negative powers of z in the expansion, $z = 0$ is a removable singularity.

$$(ii) (z + 1) \sin \frac{1}{z-2} = (t + 2 + 1) \sin \frac{1}{t} \quad \text{where } t = z - 2$$

$$\begin{aligned} &= (t + 3) \left\{ \frac{1}{t} - \frac{1}{3!t^3} + \frac{1}{5!t^5} - \dots \right\} = \left(1 - \frac{1}{3!t^2} + \frac{1}{5!t^4} - \dots \right) + \left(\frac{3}{t} - \frac{1}{2t^3} + \frac{3}{5!t^5} - \dots \right) \\ &= 1 + \frac{3}{t} - \frac{1}{6t^2} - \frac{1}{2t^3} + \frac{1}{120t^4} - \dots = 1 + \frac{3}{z-2} - \frac{1}{6(z-2)^2} - \frac{1}{2(z-2)^3} + \dots \end{aligned}$$

Since there are infinite number of terms in the negative powers of $(z - 2)$, $z = 2$ is an essential singularity.

(iii) Poles of $f(z) = \frac{1}{\cos z - \sin z}$ are given by equating the denominator to zero, i.e., by $\cos z - \sin z = 0$ or

$\tan z = 1$ or $z = \pi/4$. Clearly $z = \pi/4$ is a simple pole of $f(z)$.

Example 20.31. What type of singularity have the following functions :

$$(i) \frac{1}{1-e^z} \quad (ii) \frac{e^{2z}}{(z-1)^4} \quad (iii) \frac{e^{1/z}}{z^2}. \quad (U.P.T.U., 2009)$$

Solution. (i) Poles of $f(z) = 1/(1-e^z)$ are found by equating to zero $1-e^z = 0$ or $e^z = 1 = e^{2n\pi i}$

$$\therefore z = 2n\pi i \quad (n = 0, \pm 1, \pm 2, \dots)$$

Clearly $f(z)$ has a simple pole at $z = 2n\pi i$.

$$\begin{aligned} (ii) \frac{e^{2z}}{(z-1)^4} &= \frac{e^{2(t+1)}}{t^4} = \frac{e^2}{t^4} \cdot e^{2t} \quad \text{where } t = z-1 \\ &= \frac{e^2}{t^4} \left\{ 1 + \frac{2t}{1!} + \frac{(2t)^2}{2!} + \frac{(2t)^3}{3!} + \frac{(2t)^4}{4!} + \frac{(2t)^5}{5!} + \dots \right\} = e^2 \left\{ \frac{1}{t^4} + \frac{2}{t^3} + \frac{2}{t^2} + \frac{4}{3t} + \frac{2}{3} + \frac{4t}{15} + \dots \right\} \\ &= e^2 \left\{ \frac{1}{(z-1)^4} + \frac{2}{(z-1)^3} + \frac{2}{(z-1)^2} + \frac{4}{3(z-1)} + \frac{2}{3} + \frac{4}{15}(z-1) + \dots \right\} \end{aligned}$$

Since there are finite (4) number of terms containing negative powers of $(z-1)$,

$\therefore z = 1$ is a pole of 4th order.

$$(iii) f(z) = \frac{e^{1/z}}{z^2} = \frac{1}{z^2} \left\{ 1 + \frac{1}{1!z} + \frac{1}{2!z^2} + \frac{1}{3!z^3} + \dots \right\} = z^{-2} + z^{-3} + \frac{z^{-4}}{2} + \dots \infty$$

Since there are infinite number of terms in the negative powers of z , therefore $f(z)$ has an essential singularity at $z = 0$.

Integration of complex function along a contour $\rightarrow$ A continuous chain of a finite number of regular arcs is known as contour.

Q1 Find the value of the integral $\int_0^{1+i} (z-y+ix^2) dz$,

(a) along the straight line from $z=0$ to $z=1+i$.

(b) along the real axis from $z=0$ to $z=1$ & then along a line parallel to imaginary axis from $z=1$ to $z=1+i$.

Solⁿ

$$z = x + iy$$

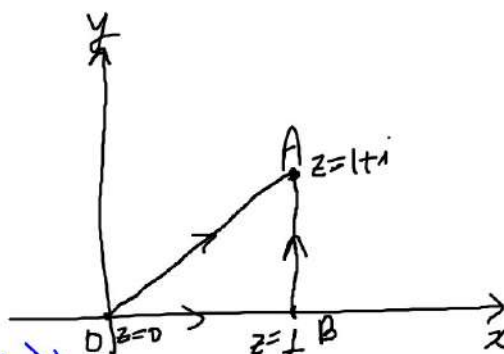
$$dz = dx + i dy$$

(a)

Along OA $\rightarrow y=x$ then $dy=dx$

$$\int_{OA} (z-y+ix^2) dz = \int_0^1 (z-x+ix^2)(dx+idx)$$

$$= (1+i)i \int_0^1 x^2 dx = (1+i)i \left[\frac{x^3}{3} \right]_0^1 = \frac{i(1+i)}{3} = \frac{i-1}{3}$$



(b) Along OB, $y=0$ so $dy=0$

$$\begin{aligned}\text{then } \int_{OB} (z-y+ix^2) dz &= \int_0^1 (x-0+ix^2)(dx+i\cdot 0) \\ &= \left[\frac{x^2}{2} + i\frac{x^3}{3} \right]_0^1 = \frac{1}{2} + \frac{i}{3}\end{aligned}$$

Now along BA $x=1$ so $dx=0$

$$\begin{aligned}\int_{BA} (z-y+ix^2)(dx+idy) &= \int_0^1 (1-y+i)(0+idy) \\ &= i \left[y - \frac{y^2}{2} + iy \right]_0^1 \\ &= i \left[1 - \frac{1}{2} + i \right] \\ &= \frac{i}{2} - 1\end{aligned}$$

Now

$$\int_{OBA} (x-y+ix^2)dz = \int_{OB} (x-y+ix^2)dz + \int_{BA} (x-y+ix^2)dz$$

$$= \frac{1}{2} + \frac{1}{3} + \frac{1}{2} - 1$$

$$= \frac{2i+3i}{6} - \frac{1}{2}$$

$$= \frac{5i}{6} - \frac{1}{2}$$

Q1 Evaluate the integral $\int_C \bar{z} dz$, where C is a straight line from $(1,0)$ to $(1,1)$.

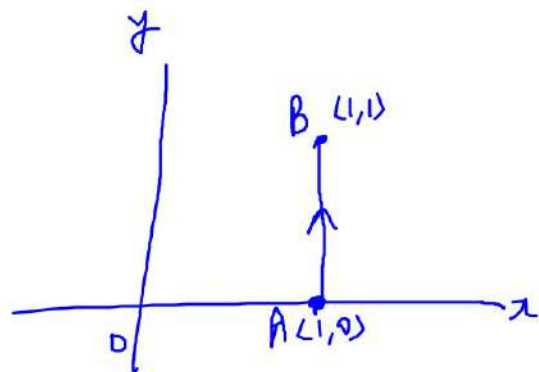
Solⁿ Along AB , $x=1 \Rightarrow dx=0$

$$\int_{AB} (x-iy)(dx+idy) = \int_0^1 (1-iy)(0+idy)$$

$$= i \left[y - i \frac{y^2}{2} \right]_0^1$$

$$= i \left[1 - \frac{i}{2} \right]$$

$$= i + \frac{1}{2}$$



Q2 Evaluate the integral (i) $\int_{-2+i}^{5+3i} z^3 dz$.

(ii) $\int_0^{1+i} z^2 dz$.

Simply & multiply connected domains → A domain in which every closed curve can be shrunk to a point without passing out of the region is called simply connected domain. If a domain is not simply connected (i.e. there is some hole), then it is called multiply connected domains.

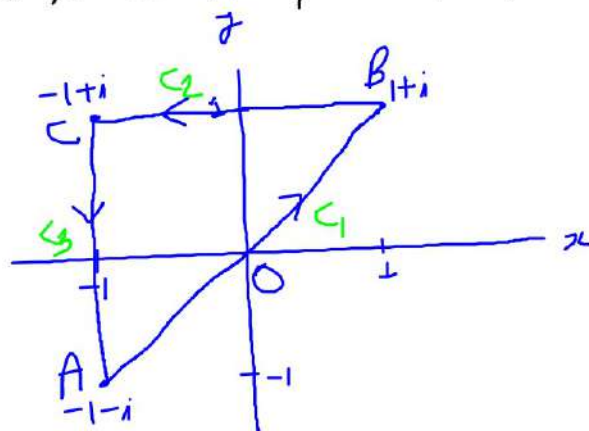
Cauchy - Goursat Theorem → If a function $f(z)$ is analytic & single valued inside & on a simple closed contour C , then

$$\int_C f(z) dz = 0.$$

Q Verify Cauchy theorem & integrating e^{iz} along the boundary of triangle with vertices at the points $1+i$, $-1+i$, $-1-i$.

Sol $\Rightarrow \therefore f(z) = e^{iz}$ is analytic within & on closed contour ΔABC & so by Cauchy theorem

$$\int_C f(z) dz = 0. \quad \text{--- (1)}$$



$$\begin{aligned} \text{Let } I &= \int_C f(z) dz = \int_{C_1} f(z) dz + \int_{C_2} f(z) dz + \int_{C_3} f(z) dz \quad \text{--- (2)} \\ &= I_1(AB) + I_2(BC) + I_3(CA). \end{aligned}$$

Along AB, $y=1 \Rightarrow dy=0$

$$\begin{aligned} I_1(AB) &= \int_{C_1} e^{i(x+iy)} (dx + i dy) = \int_{-1}^1 e^{i(x+i)} (dx + i \cdot 0) = \int_{-1}^1 e^{(i-1)x} (1+i) dx \\ &= (1+i) \left[\frac{e^{(i-1)x}}{i-1} \right]_{-1}^1 \end{aligned}$$

$$I_1 = \frac{(1+i)}{(i-1)} [e^{i-1} - \bar{e}^{i+1}] = -i(e^{i-1} - \bar{e}^{i+1})$$

Now $I_2 = \int_{\gamma_2} e^{i(x+iy)} (dx + i dy)$ Along BC, $y=1 \Rightarrow dy=0$
 $z = (x+i \cdot 1)$
 $dz = dx$

$$= \int_1^{-1} e^{i(x+i \cdot 1)} dx = \int_1^{-1} e^{ix-1} dx$$

$$= \frac{1}{e} \left[\frac{e^{ix}}{i} \right]_1^{-1} = \frac{1}{ie} [\bar{e}^i - e^i]$$

$$= \frac{i}{e} [e^i - \bar{e}^i]$$

Along CA; $x=-1 \Rightarrow dx=0$
 $z = -1 + iy$
 $dz = i dy$

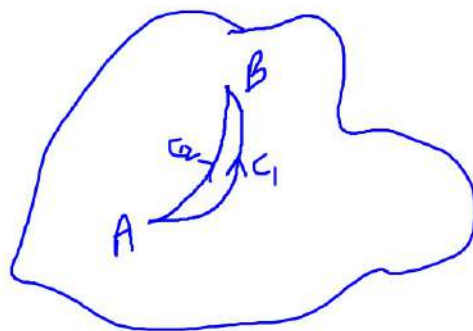
$$\begin{aligned}
 I_3(CA) &= \int_{\gamma} e^{i(x+iy)} (dx + i dy) = \int_1^{-1} e^{i(1+iy)} i dy \\
 &= i \int_1^{-1} e^{-i-y} dy \\
 &= \frac{i}{e^i} [-e^{-y}]_1^{-1} \\
 &= \frac{i}{e^i} [-e + \bar{e}] \\
 &= \frac{i}{e^i} \left[\frac{1}{e} - e \right]
 \end{aligned}$$

From eq (2),

$$\begin{aligned}
 I &= I_1 + I_2 + I_3 \\
 &= -i [e^{i-1} - \bar{e}^{i+1}] + \frac{i}{e} [e^i - \bar{e}^i] + \frac{i}{e^i} \left[\frac{1}{e} - e \right] \\
 &= \cancel{-i \frac{e^i}{e}} + i \bar{e}^i e + \cancel{i \frac{e^i}{e}} - \cancel{i \frac{\bar{e}^i}{e}} + \cancel{i \frac{\bar{e}^i}{e}} - \cancel{i \bar{e} e} \\
 I &= 0
 \end{aligned}$$

Corollary → Let $f(z)$ is analytic in a simply connected domain D then the integral along any rectifiable curve in D joining any given points of D is same, it does not depend upon the curve joining points.

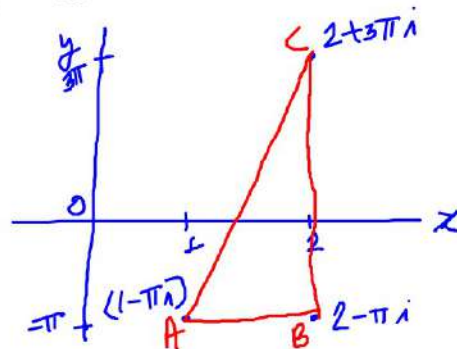
i.e.,
$$\int_{C_1} f(z) dz = \int_{C_2} f(z) dz.$$



Q → Show that $\int_C e^{-z} dz$ is independent of path C joining the points $1 - \pi i$ & $2 + 3\pi i$ & determine its value.

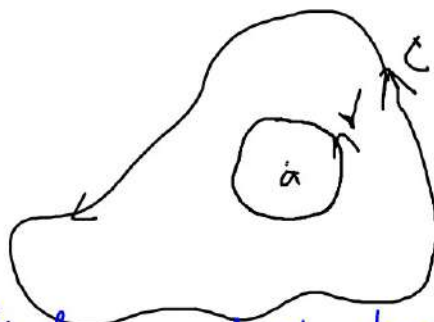
Solⁿ

$$\int_{AC} f(z) dz = \int_{AB} f(z) dz + \int_{BC} f(z) dz.$$



Cauchy integral formula $\Rightarrow$ If $f(z)$ is analytic within & on a closed contour C & ' a ' is any point within C , then

$$f(a) = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z-a)} dz$$



Remarks If a function $f(z)$ is analytic within & on a closed contour C & ' a ' is any point within C then derivatives of all orders are analytic & are given by

$$f^{(n)}(a) = \frac{n!}{2\pi i} \int_C \frac{f(z)}{(z-a)^{n+1}} dz.$$

Qn Evaluate $\int_C \frac{e^{2z}}{(z+1)^4} dz$, where C is $|z|=3$.

Soln $\therefore f^{(n)}(a) = \frac{n!}{2\pi i} \int_C \frac{f(z)}{(z-a)^{n+1}} dz$

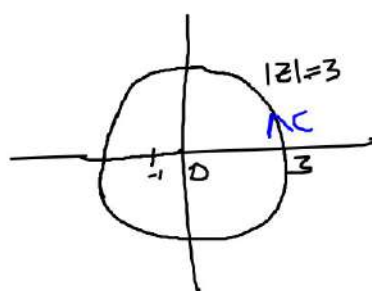
Putting $n=3$ & $a=-1$.

$$f^{(3)}(-1) = \frac{3!}{2\pi i} \int_C \frac{f(z)}{(z+1)^{3+1}} dz.$$

$$\text{so } \int_C \frac{f(z)}{(z+1)^4} dz = \frac{2\pi i}{3!} f^{(3)}(-1). \quad \text{--- (1)}$$

Take $f(z) = e^{2z}$, so $f'(z) = 2e^{2z}$, $f''(z) = 4e^{2z}$
& $f'''(z) = 8e^{2z}$

$$\text{so } f'''(-1) = 8e^{-2} \text{ i.e. } f^{(3)}(-1) = \frac{8}{e^2}$$



From eq. ①,

$$\int_C \frac{e^{2z}}{(z+1)^4} dz = \frac{2\pi i}{3!} \cdot \frac{8}{e^2}$$

$$= \frac{8\pi i}{3e^2}$$

Q1) Using Cauchy integral formula, calculate the following integrals:

(i) $\int_C \frac{z dz}{(9-z^2)(z+i)}$; where C is the circle $|z|=2$ described in +ve sense.

(ii) $\int_C \frac{dz}{z(z+\pi i)}$; where C is $|z+3i|=1$

(iii) $\int_C \frac{(\cos t) f(tz) dz}{z(z^2+1)}$; where C is $|z|=2$.

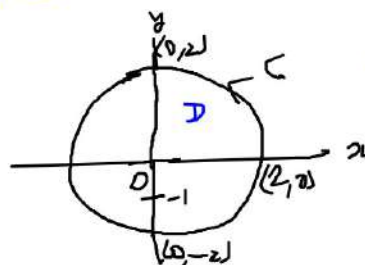
(iv) $\int_C \frac{e^{az} dz}{(z-\pi i)}$, where C is ellipse $|z-2|+|z+2|=6$.

Solⁿ, W.K.T.

$$f(a) = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z-a)} dz.$$

(i) Let $I = \int_C \frac{z dz}{(9-z^2)(z+i)}$; $D: |z|=2$

when $9-z^2=0 \Rightarrow z=\pm 3 \notin D$
& $z+i=0$ or $z=-i \in D$.



Then we take $f(z) = \frac{z}{9-z^2}$.

$$\begin{aligned} \text{Now } I &= \int_C \frac{f(z)}{(z-i)} dz = 2\pi i f(-i) \\ &= 2\pi i \times \left(\frac{-i}{9-i^2} \right) \\ &= \frac{2\pi}{10} \\ &= \pi/5. \end{aligned}$$

(ii) $I = \int_C \frac{dz}{z(z+\pi i)}$; $|z+3i|=1$.

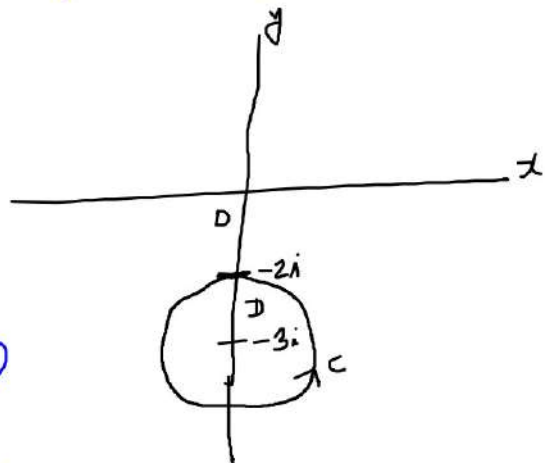
$\therefore z=0 \notin D$ & $z=-\pi i \in D$

so $f(z) = \frac{1}{z}$.

$$I = \int_C \frac{f(z)}{(z-(-\pi i))} dz = 2\pi i f(-\pi i)$$

$$= 2\pi i \times \frac{-1}{\pi i}$$

$I = -2$.

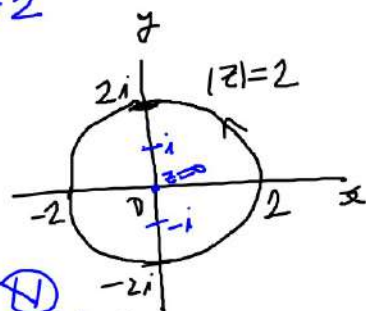


(iii) $I = \int_C \frac{\cosh(\pi z)}{z(z^2+1)} dz$, $|z|=2$

$\therefore z=0, \pm i \in D$.

$$I = \int_C \left[\frac{A}{z} + \frac{B}{z+i} + \frac{C}{z-i} \right] f(z) dz$$

where $f(z) = \cosh(\pi z)$



$$\frac{1}{z(z+i)(z-i)} = \frac{A}{z} + \frac{B}{z+i} + \frac{C}{z-i};$$

where $A = \frac{1}{(z+i)(z-i)} = \frac{1}{(0+i)(0-i)} = 1$ at $z=0$

$$B = \frac{1}{z(z-i)} = \frac{1}{(-i)(-i-i)} = -\frac{1}{2} \text{ at } z=-i$$

& $C = \frac{1}{z(z+i)} = \frac{1}{i(i+i)} = -\frac{1}{2} \text{ at } z=i.$

From eq ①,

$$I = 2\pi i [A f(0) + B f(-i) + C f(i)]$$

$$= 2\pi i \left[1 \cdot 1 - \frac{1}{2} f(-i) - \frac{1}{2} f(i) \right]$$

$$= 2\pi i \left[1 + \frac{1}{2} + \frac{1}{2} \right]$$

$$= 4\pi i$$

$$\begin{aligned} \because \operatorname{Gr} h x &= \operatorname{Gr} i x \\ f(-i) &= \operatorname{Gr} h(-\pi i) \\ &= \operatorname{Gr}(-i^2 \pi) \\ &= \operatorname{Gr} \pi \\ &= -1 \\ \text{Similarly } f(i) &= -1 \end{aligned}$$

$$\text{iv) } \therefore |z-2| + |z+2| = 6$$

$$\text{or } |(x-2)+iy| + |(x+2)+iy| = 6$$

$$\text{or } \sqrt{(x-2)^2+y^2} + \sqrt{(x+2)^2+y^2} = 6$$

$$\text{or } \sqrt{(x-2)^2+y^2} = 6 - \sqrt{(x+2)^2+y^2}$$

On squaring

$$(x-2)^2+y^2 = 36 + (x+2)^2+y^2 - 2 \times 6 \times \sqrt{(x+2)^2+y^2}$$

$$x^2+4-4x+y^2 = 36 + x^2+4+4x+y^2 - 12 \sqrt{(x+2)^2+y^2}$$

$$12 \sqrt{(x+2)^2+y^2} = 8x+36$$

$$3 \sqrt{(x+2)^2+y^2} = 2x+9$$

On squaring again

$$9[(x+2)^2+y^2] = 4x^2+81+36x$$

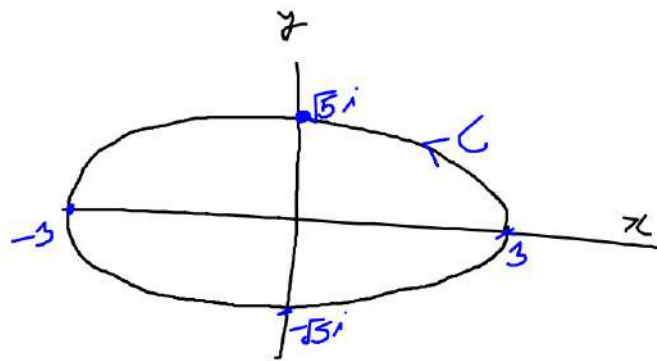
$$\text{or } 9(x^2+4+4x+y^2) = 4x^2+81+36x$$

$$\text{or } 9x^2+36+36x+9y^2 = 4x^2+81+36x$$

$$\text{or } 5x^2+9y^2=45$$

$$\text{or } \frac{x^2}{9} + \frac{y^2}{5} = 1$$

$$\text{or } \frac{x^2}{3^2} + \frac{y^2}{(\sqrt{5})^2} = 1.$$



$$\therefore I = \int \frac{e^{az}}{z-\pi i} dz ; \quad z=\pi i \notin D \text{ \& have } f(z) = \frac{e^{az}}{z-\pi i}$$

so from Cauchy theorem $I=0$. is analytic within & on C

Residue at a pole $\rightarrow$ From Laurent's series, if $f(z)$ has a pole at $z=a$ of order m

then $f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=1}^m a_{-n} (z-a)^{-n}$. ——— ①

where $a_n = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z-a)^{n+1}} dz$,

then $a_{-1} = \frac{1}{2\pi i} \int_C f(z) dz$. ——— ②

is called residue at pole $z=a$ & denoted by $\text{Res}(z=a) = a_{-1}$.

Remark ① Consider the case in which $z=a$ is a simple pole, then eq ① becomes

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \frac{a_{-1}}{(z-a)}.$$

$$\Rightarrow \lim_{z \rightarrow a} (z-a) f(z) = a_{-1}.$$

Thus using ② $\text{Res}(z=a) = \lim_{z \rightarrow a} (z-a) f(z) = a_{-1} = \frac{1}{2\pi i} \int_C f(z) dz$

here C may be replaced by any closed curve containing within it no other singularity except at $z=a$.

② Residue at infinity $\rightarrow$ If $f(z)$ has an isolated singularity at $z=a$ or is analytic there, then residue at $z=\infty$ is defined

as:

$$\text{Res}(z=\infty) = -\frac{1}{2\pi i} \int_C f(z) dz.$$

where C is any closed contour which encloses all finite singularities of $f(z)$. (Integral is taken in anticlockwise direction).

$$\text{Res}(z=\infty) = \lim_{z \rightarrow \infty} -z f(z) \text{ if limit exists.}$$

$$= -\text{ve coeff. of } \frac{1}{z} \text{ in expansion of } f(z) \text{ in neighbourhood of } z=\infty.$$

③ If $f(z) = \frac{\phi(z)}{\psi(z)}$ has a simple pole at $z=a$, then

$$\text{Res}(f(a)) = \frac{\phi(a)}{\psi'(a)} ; \text{ which is applied where } \psi(z) \text{ can not be factorised.}$$

④ If $f(z) = \frac{\phi(z)}{(z-a)^m}$ has a pole of order m then

$$\operatorname{Res}[f(z=a)] = \frac{\phi^{(m-1)}(a)}{(m-1)!}$$

Cauchy's residue theorem $\Rightarrow$ If $f(z)$ is analytic within & on a closed contour C , except at a finite number of poles $z_1, z_2, \dots, z_n$ within C , then

$$\boxed{\int_C f(z) dz = 2\pi i \sum_{r=1}^n \text{Res}(z=z_r)}$$

where R.H.S denotes sum of residues of $f(z)$ at its poles lying within C .

Remark $\Rightarrow$ If a function $f(z)$ is analytic except at finite number of singularities (including that at infinity), then the sum of residues of these singularities is zero.

Q3 Evaluate the residues of $\frac{z^2}{(z-1)(z-2)(z-3)}$ at 1, 2, 3 & ∞ .

Also show that their sum is zero.

Solⁿ Let $f(z) = \frac{z^2}{(z-1)(z-2)(z-3)}$.

$\therefore z=1, 2, 3$ are the simple poles of $f(z)$.

$$\begin{aligned} \text{Res}(z=1) &= \lim_{z \rightarrow 1} (z-1) f(z) = \lim_{z \rightarrow 1} (z-1) \frac{z^2}{(z-1)(z-2)(z-3)} \\ &= \frac{1}{(1-2)(1-3)} = \frac{1}{2}. \end{aligned}$$

$$\text{Res}(z=2) = \lim_{z \rightarrow 2} (z-2) f(z) = \lim_{z \rightarrow 2} \frac{z^2}{(z-1)(z-3)} = -4.$$

$$\text{Res}(z=3) = \lim_{z \rightarrow 3} (z-3) f(z) = \lim_{z \rightarrow 3} \frac{z^2}{(z-1)(z-2)} = \frac{9}{2}$$

$$\begin{aligned} \text{Res}(z=+\infty) &= \lim_{z \rightarrow \infty} -z f(z) = \lim_{z \rightarrow \infty} \frac{-z^3}{(z-1)(z-2)(z-3)} = \lim_{z \rightarrow \infty} \frac{-1}{(1-\frac{1}{z})(1-\frac{2}{z})(1-\frac{3}{z})} = -1 \\ &\quad \downarrow \\ &\quad \text{when limit exists} \end{aligned}$$

$$\text{Sum of residues} = \frac{1}{2} - 4 + \frac{9}{2} - 1$$

$$= 0.$$

Qs Evaluate the residue of $\frac{z^3}{(z-1)(z-2)(z-3)}$ at $z=\infty$.

Solⁿ Let $f(z) = \left(1 - \frac{1}{z}\right)^{-1} \left(1 - \frac{2}{z}\right)^{-1} \left(1 - \frac{3}{z}\right)^{-1}$

$$= \left(1 + \frac{1}{z} + \dots\right) \left(1 + \frac{2}{z} + \dots\right) \left(1 + \frac{3}{z} + \dots\right)$$

$$= \left(1 + \frac{2}{z} + \frac{1}{z} + \dots\right) \left(1 + \frac{3}{z} + \dots\right)$$

$$= \left(1 + \frac{3}{z} + \dots\right) \left(1 + \frac{3}{z} + \dots\right)$$

$$= 1 + \frac{3}{z} + \frac{3}{z} + \dots$$

$$= 1 + \frac{6}{z} + \text{higher powers of } \frac{1}{z}$$

$$\text{Res}(f(\infty)) = -\text{ve coeff of } \frac{1}{z} = -6.$$

Example 20.33. Determine the poles of the function

$$f(z) = z^2/(z-1)^2(z+2) \text{ and the residue at each pole.} \quad (\text{S.V.T.U., 2008 ; J.N.T.U., 2005})$$

Hence evaluate $\oint_C f(z)dz$, where C is the circle $|z| = 2.5$.

Solution. Since $\lim_{z \rightarrow -2} [(z+2)f(z)] = \lim_{z \rightarrow -2} \frac{z^2}{(z-1)^2} = \frac{4}{9}$,

which is finite and non-zero, the function has a simple pole at $z = -2$ and $\text{Res } f(-2) = 4/9$.

Also since $\lim_{z \rightarrow 1} [(z-1)^2 f(z)]$ is finite and non-zero, $f(z)$ has a pole of order two at $z = 1$.

$$\therefore \text{Res } f(1) = \frac{1}{1!} \left[\frac{d}{dz} [(z-1)^2 f(z)] \right]_{z=1} = \left[\frac{d}{dz} \left(\frac{z^2}{z+2} \right) \right]_{z=1} = \left[\frac{z^2 + 4z}{(z+2)^2} \right]_{z=1} = \frac{5}{9}.$$

[Otherwise writing $z = 1 + t$,

$$\begin{aligned} f(z) &= \frac{(1+t)^2}{t^2(3+t)} = \frac{1}{3t^2} (1+t)^2 (1+t/3)^{-1} = \frac{1}{3t^2} (1+t)^2 \left(1 - \frac{t}{3} + \frac{t^2}{9} - \dots \right) \\ &= \frac{1}{3t^2} \left(1 + \frac{5}{3}t + \frac{4}{9}t^2 - \dots \right) = \frac{1}{3t^2} + \frac{5}{9t} + \frac{4}{27} - \dots \end{aligned} \quad \dots(i)$$

$$\therefore \text{Res } f(1) = \text{coefficient of } \frac{1}{t} \text{ in (i)} = \frac{5}{9}.$$

Clearly $f(z)$ is analytic on $|z| = 2.5$ and at all points inside except the poles $z = -2$ and $z = 1$. Hence by residue theorem

$$\oint_C f(z)dz = 2\pi i [\text{Res } f(-2) + \text{Res } f(1)] = 2\pi i \left[\frac{4}{9} + \frac{5}{9} \right] = 2\pi i.$$

Example 20.35. Evaluate

$$\oint_C \frac{z-3}{z^2+2z+5} dz, \text{ where } C \text{ is the circle}$$

$$(i) |z| = 1, \quad (ii) |z+1-i| = 2, \quad (iii) |z+1+i| = 2. \quad (J.N.T.U., 2003)$$

Solution. The poles of $f(z) = \frac{z-3}{z^2+2z+5}$ are given by $z^2+2z+5=0$

i.e., by
$$z = \frac{-2 \pm \sqrt{4-20}}{2} = -1 \pm 2i.$$

(i) Both the poles $z = -1+2i$ and $z = -1-2i$ lie outside the circle $|z| = 1$. Therefore, $f(z)$ is analytic everywhere within C .

Hence by Cauchy's theorem,
$$\oint_C \frac{z-3}{z^2+2z+5} dz = 0.$$

(ii) Here only one pole $z = -1+2i$ lies inside the circle $C : |z+1-i| = 2$. Therefore, $f(z)$ is analytic within C except at this pole.

$$\begin{aligned} \therefore \text{Res } f(-1+2i) &= \lim_{z \rightarrow -1+2i} [(z - (-1+2i)) f(z)] = \lim_{z \rightarrow -1+2i} \frac{(z+1-2i)(z-3)}{z^2+2z+5} \\ &= \lim_{z \rightarrow -1+2i} \frac{z-3}{z+1+2i} = \frac{-4+2i}{4i} = i + 1/2. \end{aligned}$$

Hence by residue theorem
$$\oint_C f(z) dz = 2\pi i \text{ Res } f(-1+2i) = 2\pi i(i + 1/2) = \pi(i-2).$$

(iii) Here only the pole $z = -1-2i$ lies inside the circle $C : |z+1+i| = 2$. Therefore, $f(z)$ is analytic within C except at this pole.

$$\begin{aligned} \therefore \text{Res } f(-1-2i) &= \lim_{z \rightarrow -1-2i} \frac{(z+1+2i)(z-3)}{z^2+2z+5} \\ &= \lim_{z \rightarrow -1-2i} \frac{z-3}{z+1-2i} = \frac{-4-2i}{-4i} = \frac{1}{2} - i \end{aligned}$$

Hence by residue theorem,
$$\oint_C f(z) dz = 2\pi i \text{ Res } f(-1-2i) = 2\pi i(\frac{1}{2} - i) = \pi(2+i).$$

Example 20.37. Evaluate $\oint_C \tan z \, dz$ where C is the circle $|z| = 2$.

(V.T.U., 2010 S)

Solution. The poles of $f(z) = \sin z / \cos z$ are given by $\cos z = 0$ i.e. $z = \pm \pi/2, \pm 3\pi/2, \pm 5\pi/2, \dots$
Of these poles, $z = \pi/2$, and $-\pi/2$ only are within the given circle.

$$\therefore \text{Res } f(\pi/2) = \lim_{z \rightarrow \pi/2} \frac{\sin z}{\frac{d}{dz}(\cos z)} = \lim_{z \rightarrow \pi/2} \left(\frac{\sin z}{-\sin z} \right) = -1 \quad [\text{By } \S 20.19 (2)]$$

$$\text{Similarly } \text{Res } f(-\pi/2) = \lim_{z \rightarrow -\pi/2} \frac{\sin z}{\frac{d}{dz}(\cos z)} = -1.$$

Hence by residue theorem,

$$\oint_C f(z) dz = 2\pi i [\text{Res } f(\pi/2) + \text{Res } f(-\pi/2)] = 2\pi i (-1 - 1) = -4\pi i.$$

Example 20.38. Evaluate $\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2(z-2)} dz$, where C is the circle $|z| = 3$.

(V.T.U., 2010 ; Anna, 2003 S ; U.P.T.U., 2002)

$$\text{Solution. } f(z) = \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2(z-2)}$$

is analytic within the circle $|z| = 3$ excepting the poles $z = 1$ and $z = 2$.

Since $z = 1$ is a pole of order 2.

$$\begin{aligned} \therefore \text{Res } f(1) &= \frac{1}{1!} \left[\frac{d}{dz} \{(z-1)^2 f(z)\} \right]_{z=1} = \left[\frac{d}{dz} \left(\frac{\sin \pi z^2 + \cos \pi z^2}{(z-2)} \right) \right]_{z=1} \\ &= \left[\frac{(z-2)(2\pi z \cos \pi z^2 - 2\pi z \sin \pi z^2) - (\sin \pi z^2 + \cos \pi z^2)}{(z-2)^2} \right]_{z=1} \\ &= (-1)(-2\pi) - (-1) = 2\pi + 1 \end{aligned}$$

$$\text{Also } \text{Res } f(2) = \lim_{z \rightarrow 2} [(z-2) f(z)] = \lim_{z \rightarrow 2} \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2} = 1$$

Hence by residue theorem,

$$\oint_C f(z) dz = 2\pi i [\text{Res } f(1) + \text{Res } f(2)] = 2\pi i (2\pi + 1 + 1) = 4\pi(\pi + 1)i.$$

20.20 EVALUATION OF REAL DEFINITE INTEGRALS

Many important definite integrals can be evaluated by applying the Residue theorem to properly chosen integrals. The contours chosen will consist of straight lines and circular arcs.

(a) Integration around the unit circle. An integral of the type $\int_0^{2\pi} f(\sin \theta, \cos \theta) d\theta$, where the integrand is a rational function of $\sin \theta$ and $\cos \theta$ can be evaluated by writing $e^{i\theta} = z$.

Since $\sin \theta = \frac{1}{2i} \left(z - \frac{1}{z} \right)$ and $\cos \theta = \frac{1}{2} \left(z + \frac{1}{z} \right)$, then integral takes the form $\int_C f(z) dz$, where $f(z)$ is a rational function of z and C is a unit circle $|z| = 1$.

Hence the integral is equal to $2\pi i$ times the sum of the residues at those poles of $f(z)$ which are within C .

$$|z|=1 \Rightarrow z=e^{i\theta}$$

$$z = e^{i\theta} = \cos \theta + i \sin \theta \quad \text{--- ①}$$

$$\frac{1}{z} = e^{-i\theta} = \cos \theta - i \sin \theta \quad \text{--- ②}$$

$$\text{①} + \text{②}; \quad \cos \theta = \frac{1}{2} \left(z + \frac{1}{z} \right).$$

$$\text{①} - \text{②}; \quad \sin \theta = \frac{1}{2i} \left(z - \frac{1}{z} \right).$$

Q_n prove that $\int_0^{2\pi} \frac{\cos 2\theta \, d\theta}{1 - 2a \cos \theta + a^2} = \frac{2\pi a^2}{1 - a^2}; \quad a^2 < 1.$

Solⁿ $\because |z|=1 \Rightarrow z = e^{i\theta}$ then $dz = i e^{i\theta} d\theta$
or $d\theta = \frac{dz}{i z}$

Putting $\cos \theta = \frac{1}{2} \left(z + \frac{1}{z} \right)$ & $\sin \theta = \frac{1}{2i} \left(z - \frac{1}{z} \right)$

so $\cos 2\theta = \frac{1}{2} \left(z^2 + \frac{1}{z^2} \right)$

From the integral $\int_0^{2\pi} \frac{\cos 2\theta}{1 - 2a \cos \theta + a^2} d\theta = \frac{1}{2} \int_C \frac{\left(z^2 + \frac{1}{z^2} \right) \frac{dz}{i z}}{1 - 2a \times \frac{1}{2} \left(z + \frac{1}{z} \right) + a^2}$

$$= \frac{1}{2i} \int_C \frac{(z^4 + 1)}{z^2 [z - az^2 - a + az]} dz$$

$$= \frac{1}{2i} \int_C \frac{(z^4 + 1)}{z^2 [z(1 - az) - a(1 - az)]} dz$$

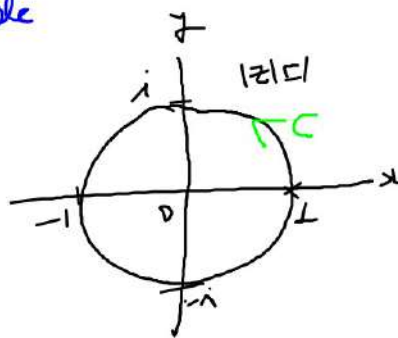
$$= \frac{1}{2i} \int_C \frac{(z^4+1)}{z^2(z-a)(1-az)} dz ;$$

$$I = \int_C f(z) dz \quad ; \text{ where } C \text{ is unit circle i.e. } |z|=1$$

where $f(z) = \frac{z^4 + 1}{2i [z^2 (z-a) (1-az)]}$ ————— ①

Poles of $f(z)$ are: $z=0$, $z=a$ & $z=\frac{1}{a}$
 $\downarrow$ $\downarrow$
of order 2 simple pole

Given $a^2 < 1$.

$$z=0, a \in \mathbb{C}.$$
$$z = \frac{1}{a} \notin C.$$


$$\text{Now } \text{Res}(z=0) = \frac{\phi^{(2-1)}(0)}{(2-1)!}$$

of order 2

$$f(z) = \frac{\phi(z)}{z^2},$$

$$= \frac{\phi'(0)}{1!} \quad \text{--- ②}$$

$$\left\{ \begin{aligned} f(z) &= \frac{\phi(z)}{(z-a)^m} \\ \text{Res}(z=a) &= \frac{\phi^{(m-1)}(a)}{(m-1)!} \end{aligned} \right.$$

$$\text{where } \phi(z) = \frac{z^4+1}{2i(z-a)(1-az)} = \frac{z^4+1}{2i[z-az^2-a+az^2]} \quad \text{--- ②}$$

$$\phi'(z) = \frac{1}{2i} \frac{(z-az^2-a+az^2)(4z^3) - (z^4+1)(1-2az+az^2)}{(z-az^2-a+az^2)^2}$$

$$\phi'(0) = \frac{1}{2i} \frac{0 - (0+1)(1-0+a^2)}{a^2}$$

$$\phi'(0) = -\frac{1}{2i} \left(\frac{1+a^2}{a^2} \right)$$

From ②

$$\text{Res}(z=0) = -\frac{1}{2i} \left(\frac{1+a^2}{a^2} \right) \quad \text{--- ②}$$

Now $\text{Res}(z=a) = \lim_{z \rightarrow a} (z-a) f(z)$

←
simple pole

$$= \lim_{z \rightarrow a} 2i \frac{z^4 + 1}{[z^2(1-az)]}$$

$$\text{Res}(z=a) = \frac{a^4 + 1}{2i \times a^2(1-a^2)} \quad \text{--- (3)}$$

From Cauchy's residue theorem:

$$\int_C f(z) dz = 2\pi i [\text{Res}(z=0) + \text{Res}(z=a)]$$

$$= 2\pi i \left[-\frac{(1+a^2)}{2ia^2} + \frac{a^4+1}{2ia^2(1-a^2)} \right]$$

$$= \frac{2\pi i}{2ia^2} \left[\frac{-(1-a^4) + a^4 + 1}{1-a^2} \right] = \frac{2\pi a^2}{1-a^2}$$

Q.15 By integrating around a unit circle, evaluate

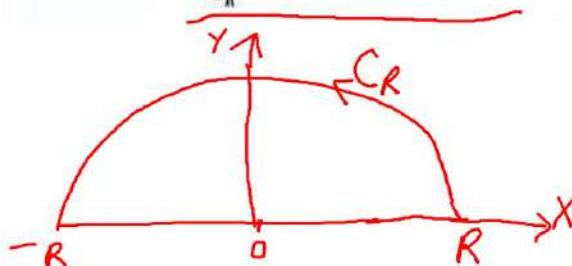
$$\int_0^{2\pi} \frac{\cos 3\theta}{5 - 4 \cos \theta} d\theta.$$

(b) **Integration around a small semi-circle.** To evaluate $\int_{-\infty}^{\infty} f(x) dx$, we consider $\int_C f(z)$, where C is i.e. poles are not lying on real axis $\Rightarrow$ the contour consisting of the semi-circle $C_R: |z| = R$, together with the diameter that closes it.

Supposing that $f(z)$ has no singular point on the real axis, we have, by the Residue theorem,

$$\left[\int_{C_R} f(z) dz + \int_{-R}^R f(x) dx = 2\pi i \sum \text{Res } f(a). \right]$$

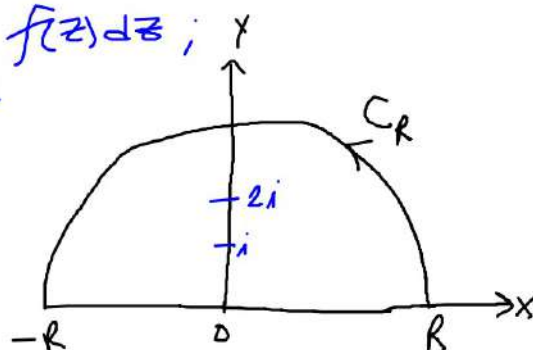
Finally making R tend to ∞ , we find the value of $\int_{-\infty}^{\infty} f(x) dx$, provided $\int_{C_R} f(z) dz \rightarrow 0$ as $R \rightarrow \infty$.



Q1 Evaluate $\int_{-\infty}^{\infty} \frac{x^2 dx}{(x^2+1)(x^2+4)}$.

Solⁿ Let $\int_C \frac{z^2 dz}{(z^2+1)(z^2+4)} = \int_C f(z) dz;$

where $f(z) = \frac{z^2}{(z^2+1)(z^2+4)}$



Poles of $f(z)$ are $z = \pm i, \pm 2i$ of order 1.

So $i, 2i \in C$ & $-i, -2i \notin C$.

From Cauchy-Residue theorem:

$$\int_C f(z) dz = 2\pi i \left[\text{Res}(z=i) + \text{Res}(z=2i) \right]$$

$$= 2\pi i \left[\lim_{z \rightarrow i} (z-i) \frac{z^2}{(z+1)(z-i)(z^2+4)} + \lim_{z \rightarrow 2i} (z-2i) \frac{z^2}{(z^2+1)(z+2i)(z-2i)} \right]$$

$$\int_C f(z) dz = 2\pi i \left[\frac{i^2}{2i(i^2+4)} + \frac{4i^2}{(4i^2+1)4i} \right] = 2\pi i \left[\frac{-1}{6i} + \frac{1}{3i} \right] = \frac{\pi}{3} \quad \text{--- (1)}$$

$$\therefore \int_C f(z) dz = \int_{-R}^R f(x) dx + \int_{C_R} f(z) dz$$

When $R \rightarrow \infty$, we can show that $\int_{C_R} f(z) dz \rightarrow 0$. For any point C_R on $|z| \rightarrow \infty$,

$$f(z) = \frac{1}{z(1+\frac{1}{z})(1+\frac{4}{z^2})} \quad \text{decreases as } \frac{1}{z^2}$$

& $\rightarrow 0$ whereas the length of C_R increases with z .

$$\text{Consequently } \lim_{|z| \rightarrow \infty} \int_{C_R} f(z) dz = 0. \quad \text{--- (3)}$$

So from eq (1), (2) & (3)

$$\int_{-\infty}^{\infty} \frac{x^2 dx}{(x^2+1)(x^2+4)} = \frac{\pi}{3}$$

Ans.

Q's Evaluate $\int_0^{\infty} \frac{\cos x}{(x^2+1)} dx$.

(c) Indenting the contours having poles on real axis $\rightarrow$
 $\leftarrow$

Results ① If AB is the arc $\theta_1 \leq \theta \leq \theta_2$ of the circle $|z-a|=\gamma$ & if $\lim_{z \rightarrow a} (z-a)f(z) = k$ (constant),

then
$$\lim_{\gamma \rightarrow 0} \int_{AB} f(z) dz = i(\theta_2 - \theta_1)k.$$

② If $0 \leq \theta \leq \frac{\pi}{2}$, then the inequality $\frac{2\theta}{\pi} \leq \sin \theta \leq \theta$ is known as Jordan's inequality.

③ Jordan's Lemma $\rightarrow$ If $f(z)$ is analytic except at finite number of singularities & if $f(z) \rightarrow 0$ uniformly as $z \rightarrow \infty$, then

$$\lim_{R \rightarrow \infty} \int_{C_R} e^{imz} f(z) dz = 0, \quad m > 0,$$

where C_R denotes the semi-circle $|z|=R$; $\text{Im}(z) > 0$.

Q₁ Apply the calculus of residues to prove that

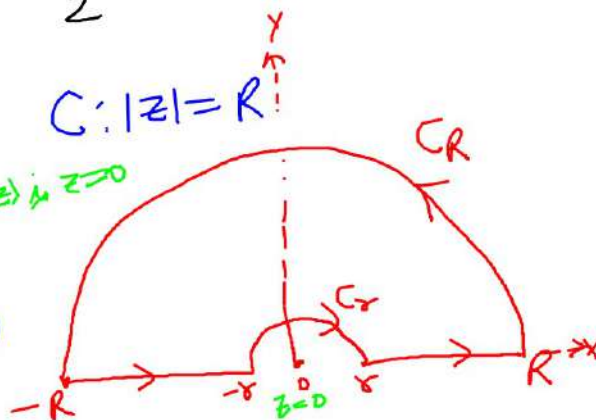
$$\int_0^{\infty} \frac{\sin mx}{x} dx = \frac{\pi}{2}, \text{ when } m > 0.$$

Solⁿ Let $f(z) = \frac{e^{imz}}{z}$; $C: |z| = R$

From Cauchy's residue theorem: pole of $f(z)$ is $z=0$

$$\int_C f(z) dz = 2\pi i \times (\text{Sum of residues within } C)$$

$$\int_C f(z) dz = 0$$



$$\int_C f(z) dz = \int_r^R f(x) dx + \int_{C_R} f(z) dz + \int_{-R}^{-r} f(x) dx + \int_{C_r} f(z) dz$$

i.e.

$$\int_r^R f(x) dx + \int_{C_R} f(z) dz + \int_{-R}^{-r} f(x) dx + \int_{C_r} f(z) dz = 0 \quad \text{--- (1)}$$

Since $\lim_{z \rightarrow 0} z f(z) = \lim_{z \rightarrow 0} z \times \frac{e^{imz}}{z} = 1.$

& $\lim_{z \rightarrow 0} \int_{C_R} f(z) dz = i(0 - \pi)1 = -i\pi$ ——— ②

By Jordan's lemma ; since $\frac{1}{z} \rightarrow 0$ in $f(z)$ as $|z| \rightarrow \infty$

then $\lim_{R \rightarrow \infty} \int_{C_R} \frac{e^{imz}}{z} dz = \lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = 0.$ — ③

Now making $\epsilon \rightarrow 0$ & $R \rightarrow \infty$ in ① & noting ② & ③

$$\int_0^{\infty} f(x) dx + \int_{-\infty}^0 f(x) dx + (-i\pi) = 0$$

$$\int_{-\infty}^{\infty} f(x) dx = i\pi \quad \text{or} \quad \int_{-\infty}^{\infty} \frac{e^{imx}}{x} dx = i\pi$$

$$\text{or } \int_{-\infty}^{\infty} \left(\frac{\cos mx + i \sin mx}{x} \right) dx = i\pi \quad \text{--- (4)}$$

equating imaginary parts on both sides, we get

$$\int_{-\infty}^{\infty} \frac{\sin mx}{x} dx = \pi$$

$$2 \int_0^{\infty} \frac{\sin mx}{x} dx = \pi$$

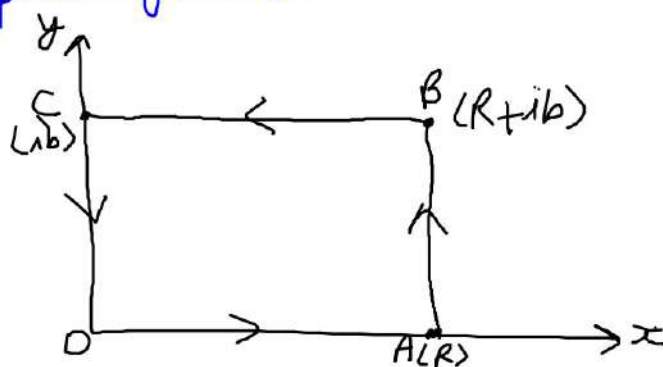
$$\text{or } \left[\int_0^{\infty} \frac{\sin mx}{x} dx = \frac{\pi}{2} \right]$$

If we equate real parts on both sides of eqⁿ (4), we get

$$\left[\int_{-\infty}^{\infty} \frac{\cos mx}{x} dx = 0 \right]$$

Qs Prove that $\int_0^{\infty} \frac{\sin \pi x}{x(1-x^2)} dx = \pi.$

(d) Integration around rectangular contours $\rightarrow$ The rectangular contour is applicable to those integrals whose integrands involve hyperbolic functions or periodic functions.



Q Evaluate $\int_{-\infty}^{\infty} \frac{e^{ax}}{e^x + 1} dx$; where $0 < a < 1$.

Solⁿ Let $\int_C \frac{e^{az}}{e^z + 1} dz = \int_C f(z) dz$; where C is the rectangle $ABCD$ with vertices at $(R, 0)$, $(R, 2\pi)$, $(-R, 2\pi)$ & $(-R, 0)$.

$$\therefore f(z) = \frac{e^{az}}{e^z + 1}$$

Poles of $f(z)$ are

$$e^z + 1 = 0$$

$$e^z = -1$$

$$z = e^{(2n+1)\pi i}$$

$$\text{or } z = (2n+1)\pi i$$

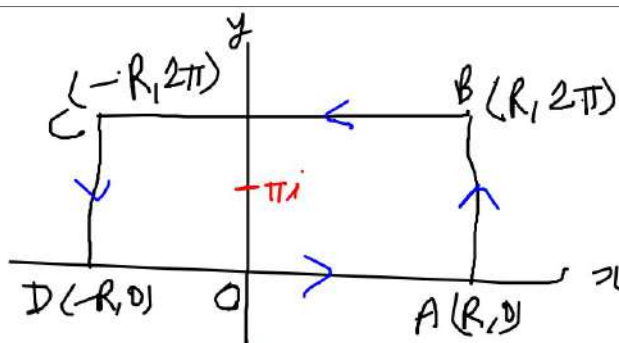
when $n=0, \pm 1, \pm 2, \dots$

for $n=0$, $z = \pi i \in \mathbb{C}$.

The only pole lies within rectangle is $z = \pi i$, which is simple pole.

From Cauchy-residue th^m:

$$\begin{aligned} \int_C f(z) dz &= 2\pi i \times \text{Res} f(z=\pi i) \\ &= 2\pi i \left[\frac{e^{az}}{\frac{d}{dz}(e^z + 1)} \right]_{z=\pi i} \end{aligned}$$



$$\begin{aligned} e^{(2n+1)\pi i} &= e^{(2n+1)\pi i + i \cdot (2n+1)\pi} \\ &= -1, \text{ for } n \in \mathbb{Z}. \end{aligned}$$

$$\begin{aligned} &\text{If } f(z) = \frac{\phi(z)}{\psi(z)}, \\ &\text{when } \psi(z) \text{ is not factorized \& has simple} \\ &\text{pole at } z=a. \\ &\text{then } \boxed{\text{Res} f(a) = \frac{\phi(a)}{\psi'(a)}} \end{aligned}$$

$$= 2\pi i \times \frac{e^{a\pi i}}{e^{\pi i}}$$

$$\int e^{\pi i} = -1$$

$$\int_C f(z) dz = -2\pi i e^{a\pi i}$$

$$\int_{AB} f(z) dz + \int_{BC} f(z) dz + \int_{CD} f(z) dz + \int_{DA} f(z) dz = -2\pi i e^{a\pi i}$$

$$\int_0^{2\pi} f(R+iy) i dy + \int_R^{-R} f(x+2\pi i) dx + \int_{2\pi}^0 f(-R+iy) i dy + \int_R^{-R} f(x) dx = -2\pi i e^{a\pi i}$$

$$i \int_0^{2\pi} \frac{e^{a(R+iy)}}{e^{R+iy} + 1} dy - \int_{-R}^R \frac{e^{a(x+2\pi i)}}{e^{x+2\pi i} + 1} dx - i \int_0^{2\pi} \frac{e^{a(-R+iy)}}{e^{-R+iy} + 1} dy$$

$$+ \int_{-R}^R \frac{e^{ax}}{e^x + 1} dx = -2\pi i e^{a\pi i}$$

$$\text{Let } i I_1 - I_2 - i I_3 + I_4 = -2\pi i e^{a\pi i} \quad \text{--- (1)}$$

For any two complex numbers z_1, z_2

if $|z_1| > |z_2|$; then $|z_1 + z_2| \geq |z_1| - |z_2|$.

so that $|e^{R+iy} + 1| \geq e^R - 1$ $|e^{R+iy}| = e^R$
 $\& |e^{R+iz}| = e^{aR}$

From I_1 , $\left| \frac{e^{a(R+iy)}}{e^{R+iy} + 1} \right| \leq \frac{e^{aR}}{e^R - 1}$ $\left| \frac{1}{e^{R+iy} + 1} \right| \leq \frac{1}{e^R - 1}$

$$= \frac{1}{e^{R(1-a)} - e^{aR}} \rightarrow 0 \text{ as } R \rightarrow \infty$$

Similarly from I_3 , $\left| \frac{e^{a(-R+iz)}}{\bar{e}^{R+iz} + 1} \right| \leq \frac{\bar{e}^{aR}}{1 - \bar{e}^R} \rightarrow 0 \text{ as } R \rightarrow \infty$

From eq ①; as $R \rightarrow \infty$; I_1 & $I_3 \rightarrow 0$, we get

$$-2\pi i \int_{-\infty}^{\infty} \frac{e^{ax}}{e^x + 1} dx + \int_{-\infty}^{\infty} \frac{e^{ax}}{e^x + 1} dx = -2\pi i e^{a\pi i}$$

$$(1 - e^{2a\pi i}) \int_{-\infty}^{\infty} \frac{e^{ax}}{e^x + 1} dx = -2\pi i e^{a\pi i}$$

$$(1 - \cos 2a\pi - i \sin 2a\pi) \int_{-\infty}^{\infty} \frac{e^{ax}}{e^x + 1} dx = -2\pi i (\cos a\pi + i \sin a\pi)$$

Equating real parts on both sides of above eqⁿ, we get

$$(1 - \cos 2a\pi) \int_{-\infty}^{\infty} \frac{e^{ax}}{e^x + 1} dx = 2\pi \sin a\pi$$

$$\int_{-\infty}^{\infty} \frac{e^{ax}}{e^x + 1} dx = \frac{2\pi \sin a\pi}{1 - 1 + 2 \sin^2 a\pi}$$

$$\int_{-\infty}^{\infty} \frac{e^{ax}}{e^x + 1} dx = \frac{\pi}{\sin a\pi}$$

Ans.

Qs Show that $\int_0^{\infty} e^{-x^2} \cos 2mx \, dx = \frac{1}{2} \sqrt{\pi} e^{-m^2}.$